Lecture 4: Evaluating and choosing

How do you score a generated image?

What would "good" even mean?

Every metric so far is unavailable

Dice, accuracy, MSE all compare a prediction to *its* ground truth. A generated image has **no counterpart** to compare against: that is the point of generating it.

Two questions instead

1. **Realism**: does the *set* of samples look like the real distribution?
2. **Utility**: does adding them improve a real task?

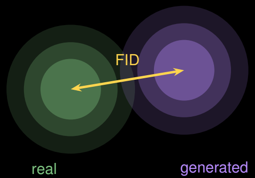
Note the shift

Both questions are about a **distribution**, not an image. You cannot score one generated scan: only a population of them.

FID, AND WHY TO DISTRUST IT HERE

Fréchet Inception Distance

Push real and generated sets through a pretrained **Inception** network, fit a Gaussian to each feature set, measure the distance between them. Lower is better.



Three problems in medical imaging

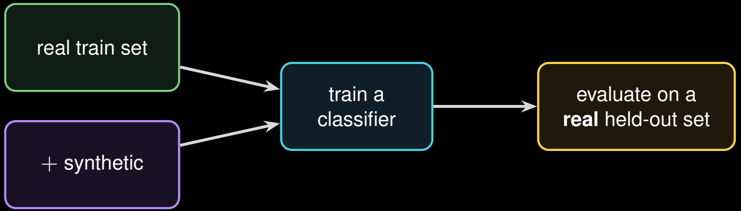
1. The features come from a network trained on **natural images**: cats and cars, not liver texture.

2. The estimate is **unstable** below a few thousand samples, and medical sets are often smaller.

3. It is blind to **anatomical correctness**. A scan with three kidneys can score beautifully.

Heusel et al., *GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium*, NeurIPS

The honest test: downstream utility



Why this is the real test

It measures the only thing you actually wanted: whether the synthetic data **helped**. If a real metric does not move, the samples did not earn their place, however good they looked.

The rule

The test set must be **real, and untouched** by the generator. Evaluating on synthetic data measures how well you imitated yourself.

Choosing, and using

Which one should you use?

	VAE	GAN	Diffusion
Training	stable	unstable	stable
Samples	blurry	sharp	sharp
Sampling	1 pass	1 pass	50–1000 passes
Latent	structured	implicit	none (pixel/latent)
Conditioning	awkward	awkward	natural

Default

Use **diffusion** unless inference cost forbids it. It is the only column with no serious weakness except compute.

Still reasons for the others

VAE when you need a structured latent: anomaly detection scores reconstruction error. **GAN** when generation must be real-time.

Pitfalls

Posterior collapse

VAE: the KL term wins, the decoder ignores z, and every sample is the dataset mean. Symptom: samples that barely vary.

Mode collapse

GAN: the rare pathology you generated the data *for* is the first mode dropped. Always check per-class coverage, not overall realism.

Memorisation: the clinical one

Diffusion models can reproduce training images **near-verbatim**, especially when a class has few examples. Two consequences:

it is a **privacy leak**, so "synthetic therefore shareable" does not follow automatically; and

synthetic anatomy can be confidently wrong, so **never diagnose on a generated image**.

Check memorisation the same way you check leakage: retrieve each sample's nearest training neighbour and look.

Hands-on: VAE & GAN on MedMNIST

Notebook

`L3_synthesis_vae_gan.ipynb`

1. Load a MedMNIST set (28–32 px).
2. Build and train a **VAE** (MSE + KL); sample a grid.
3. Build and train a small **DCGAN**; sample a grid.
4. Compare: VAE blurry, GAN sharp. See it yourself.
5. Score both with FID and note how little it tells you.

You will produce this

a grid of generated samples

VAE and GAN, side by side

VAE runs on free Colab; the GAN wants a stronger GPU.

Diffusion is left to the notebook's extension exercise: training one from scratch is a weekend, not a lab.